# The Power Rule (template)

**You missed this lesson. Copy this into your notes. That is all you need to do.**

### Question of the day

How fast does x? change, for any n?

### Targets

- I can state the power rule in words.
- I can differentiate a polynomial term by term.
- I can find the slope at a point.

### Concepts

- **Derivative** (f?(x)): The slope of the curve at a point. The rate of change of the output.
- **Power rule** (d/dx x? = n x?$^{-1}$): Bring the power down in front. Then lower the power by 1.

### What the class predicted, and what is true

- The derivative of $x^2$ is $2x$. What do you expect for $x^3$? $3x^2$. Bring the 3 down and lower the power: $3x^2$.
- What is the slope of the flat line $y = 7$? **0.** A flat line does not rise: the derivative is 0.

## Huddle notes (answers filled in)

### Definitions

- **Derivative:** The **slope** of the tangent line at a point.
- **Power rule:** Bring the **power** down, then **subtract 1** from it.

### Worked example

Differentiate f(x) = 4x$^3$.

- Bring down the power: 3 · 4x
- Lower the power by 1: 12x**2**
- So f?(x) = **12x$^2$**

### Label the slope

1. The **point** where we measure the slope.
2. The **tangent line**; its slope is f?.

# One worked sample of each skill

## Differentiate

**Level 1.** Differentiate.
**Answer:** $4x^3$ Bring down 4 and lower the power: $4x^3$.

**Level 2.** Differentiate.
**Answer:** $12x$ Multiply 6 by 2 to get 12, then lower the power: $12x$.

**Level 3.** Differentiate term by term.
**Answer:** $10x^4 + 5$ $10x^4 + 5 + 0$. The constant 1 disappears.

**Level 4.** Differentiate. Rewrite the root as a power if that helps.
**Answer:** $\frac{6}{2\sqrt{x}}$ $6x^{1/2}$ becomes $\frac{6}{2}x^{-1/2} = \dfrac{6}{2\sqrt{x}}$.

## Slope at a point

**Level 1.** Let $f(x) = x^2$. Find the slope of the curve at $x = 2$.
**Answer: 4** $f'(x) = 2x$, so $f'(2) = 4$.

**Level 2.** Let $f(x) = 4x^2$. Find the slope of the curve at $x = 3$.
**Answer: 24** $f'(x) = 8x$, so $f'(3) = 24$.

**Level 3.** Let $f(x) = 2x^4$. Find the slope of the curve at $x = 3$.
**Answer: 216** $f'(x) = 8x^3$, so $f'(3) = 216$.

**Level 4.** Let $f(x) = 4x^4$. Find the slope of the curve at $x = 1$.
**Answer: 16** $f'(x) = 16x^3$, so $f'(1) = 16$.